about writing your own version is a bad idea, because this algorithm can and need to be fine-tuned and highly optimized to exploit GPU's power.

16.3.1 The Math of Gradient Descent

If you did take a calculus class, you probably remember the notion of derivative: given a single-variable, continuous function `f(x)`, we define its derivative as the ratio between how `f` changes in response to a small change in its argument `x`; formally we write:

$$f'(x) = \frac{\partial f}{\partial x} = \lim_{\Delta x \to 0} \frac{\Delta f}{\Delta x} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

This value can be finite, infinite or even not defined for a given function and a specific value of `x`; if for a function `f` its first derivative is always defined, we call `f` a derivable function.

For derivable functions, there are formulas that allow to find the exact mathematical definition of the derivative, which in turn is going to be a function itself. For instance, if `f(x)=x`, then `f'(x)=1` (the constant function); the derivative of the quadratic function $f(x)=x^2$ is `f'(x)=2x`, and the derivative of the exponential function $f(x)=e^x$ is $f'(x)=e^x$ (yes, it's the exponential function itself!)

There are many interesting results about the geometric interpretation of function derivates[11], but we can't go through them all here.

The most important result, from our point of view, is that if we compute the value of the first derivative of a function in a given point, it tells us if the function is growing in that point, and how